

# Claim ledger

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Science entry gate: **PASS**. Tests: 466 passed, 0 failed.

| claim                                      | strength                                          | evidence                                                              | scope                                                   |
|--------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------|
| Generic functional rank is 81              | <b>certified (lower bound) + analytic (upper)</b> | 15 cells all rank 81; 81x81 minor nonzero at 6 primes                 | count is the source literature's; 126-45=81 is analytic |
| Exact equivariant bridge with left inverse | <b>certified</b>                                  | bridge_validation.json at two primes; span equalities, not dimensions | -                                                       |
| Orientation is pinned by construction      | <b>certified</b>                                  | 11 primes, all 4 residue classes, independent verifier                | -                                                       |
| Degree 4,6,10 span equality                | <b>certified</b>                                  | common-sample comparison, holdout validated                           | -                                                       |
| Degree 8 span equality                     | <b>certified</b>                                  | rank 7 both sides; tensor-word family load-bearing                    | property of the family                                  |
| dim_Q A10 = 14                             | <b>proved</b>                                     | structural: basis vectors are the coordinate system                   | -                                                       |
| dim_Q D10 = 11                             | <b>proved</b>                                     | exact rational closure; 11x11 minor below, 3 annihilators above       | seed closure                                            |
| dim_Q Q10 = 3                              | <b>proved</b>                                     | constructed: representatives independent, spanning, each needed       | -                                                       |
| dim B10 = 12                               | <b>certified at the tested primes only</b>        | solved mod p; no rational solve exists                                | 12 <= dim_Q B10 <= 14                                   |
| dim(B10 cap P10) = 1                       | <b>certified at the tested primes only</b>        | inherits B10's gap                                                    | -                                                       |
| Degree-12 equivalence                      | <b>NOT CLAIMED</b>                                | no spinor-side enumeration exists                                     | excluded                                                |

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|----------------------------------------|--------------------|---------------------------------------------|----------|
| Any physical / type<br>IIB consequence | <b>NOT CLAIMED</b> | PO-07 open: field<br>redefinitions untested | withheld |

**Words that are not licensed anywhere in the paper**

first, unique, canonical, exhaustive, previously unknown, all orders, syzygy

**Prior art**

The enumerate-evaluate-relate workflow is Elamaran, Ferko and Scarlett, PRD 114 (2026) 026016.  
The paper claims only its exact and certified realisation for the ten-dimensional self-dual five-form.